UNIT - V

E-WASTE MANAGEMENT

Electronic waste, or e-waste, refers to all items of electrical and electronic equipments and its parts that have been discarded by its owner as waste without the intent of re-use.

Sources of E-waste: E-waste sources can be numerous. Major ones are

1) Large household appliances like refrigerators/freezers, washing machines, dishwashers, televisions.
2) Small household appliances which include toasters, coffee makers, irons, hairdryers.
3) Information Technology (IT) and Telecommunications equipments namely personal computers, telephones, mobile phones, laptops, printers, scanners, photocopiers etc.
4) Lighting equipments.
5) Equipments such as fluorescent lamps.
6) Electronic or Electrical tools i.e. handheld drills, saws, screwdrivers etc.
7) Electronic toys, leisure and sport equipments.
8) Monitoring and control instruments.
9) Automatic dispensers.

Composition of E-waste:

1) Valuable metals like gold, platinum, silver and palladium.
2) Useful metals like copper, aluminium, iron etc.
3) Hazardous substances like radioactive isotopes and mercury.
4) Toxic substances like PCB's and Dioxins.
5) Plastic like High Impact Polystyrene (HIPS), Acrylonitrile Butadiene Styrene (ABS), Polycarbonate (PC), Polyphenylene oxide (PPO), etc.
6) Glass material like Cathode Ray Tube glass made up of SiO_2, CaO, Na.

A single electronic good can have one or more of the above components. For instance, a mobile phone contains more than 40 elements, base metalssuch as Copper (Cu) and Tin (Sn), special metals such as Lithium (Li), Cobalt (Co), Indium (In) and Antimony (Sb) and precious metals such as Silver (Ag), Gold (Au), and Palladium (Pd).

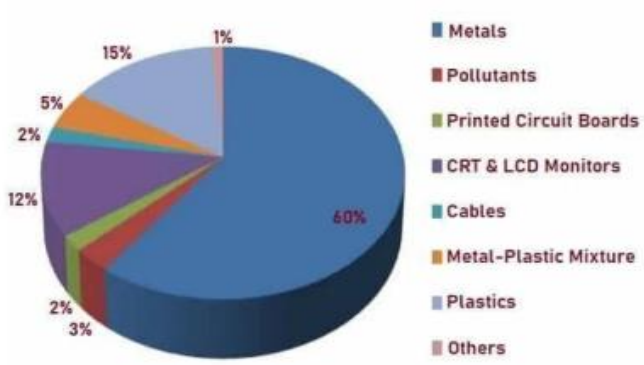

Fig. 1 Distribution of the major components found in e-wastes

Characteristics of E–waste:

E-waste includes a wide variety of items – anything from phones, computers to TVs and refrigerators. It is one of the fastest growing wastes globally. Following are its main characteristics:

1) Diverse composition: The composition of e-wastes varies in each item. It majorly contains metals (Cu, Al, etc.), plastics (PVC, polycarbonate, etc.), glass from screens and CRTS, and ceramics. Some of these materials are valuable, while some are hazardous.

(i) *Toxic substances:* E-wastes contains hazardous substances such as lead, mercury, cadmium and brominated flame retardants. These can pose serious health and environmental risks if not handled carefully.

(ii) *Non-biodegradable components:* Most of the plastics (polymers) in e-waste are non-biodegradable and persist in the environment for decades.

(iii) *Resource-rich:* E-waste is often called "urban ore" because it contains precious and rare earth metals such as platinum, gold, lithium, copper, etc. For example, one ton of e-waste may yield more gold than one ton of ore.

2) **Rapid obsolescence:** With fast technological advancements and shorter device life cycles, electronic devices are rapidly replaced by newer and more advanced models. This is leading to a rapid increase in e-waste.

3) **Disposal complexity:** Due to the complex mixture of materials tightly integrated in small devices, the recycling and safe disposal of e-wastes becomes challenging. The recycling requires manual labour for dismantling and specialized facilities for separation of hazardous and valuable components.

Need of E-waste management:

E-waste management is essential because it helps to address several environmental, health, and resource-related concerns, such as

1) Environment Protection: E-waste contains toxic substances, such as lead, mercury, and cadmium, can accumulate in the soil, air and water, if not properly managed. This can have harmful effects on the environment and human health.

2) Resource Conservation: E-waste contains valuable resources, such as metals, that can be recovered and reused through proper recycling. This helps to decrease the demand for raw materials.

3) Reducing greenhouse gas emissions: The production of new electronic products releases greenhouse gases, such as carbon dioxide, into the atmosphere. Through "reduce" and "reuse" of electronic devices, the greenhouse gas emissions can be reduced.

4) Soil Conservation: Landfilling of e-waste can result in the percolation of toxic materials into the soil. This can deplete the nutrients from soil and render it infertile.

5) Protecting public health: Improper handling and disposal of e-waste can expose workers and the general public to hazardous materials and cause serious health problems.

6) Job opportunities: Collection, dismantling, recycling and safe disposal of electronics requires skilled labor and technicians. E-waste also promotes green entrepreneurship and supports the growth of waste management industries.

Health hazards (ill effects) due to exposure to e-waste:

Due to the presence of toxic substance, exposure to e-waste can lead to serious health hazards, especially in workers handling e-waste without proper protection.

(i) **Arsenic:** Chronic exposure to arsenic can lead to various diseases of the skin and decrease nerve conduction velocity. Chronic exposure to arsenic can also cause lung cancer and can often be fatal.

(ii) **Barium**: Being highly unstable in the pure form, it forms poisonous oxides when in contact with air. Short-term exposure to barium could lead to brain swelling, muscle weakness, damage to the heart, liver and spleen. It increases blood pressure and causes heart-related diseases.

(iii) **Beryllium:** It can cause lung cancer. The primary health concern is inhalation of beryllium dust, fume or mist. Workers who are constantly exposed to beryllium, can develop beryllicosis, a disease which primarily affects the lungs. Exposure to beryllium also causes a form of skin disease that is characterized by poor wound healing and wart-like bumps. People can still develop beryllium diseases even many years following the last exposure.

(iv) **Brominated flame retardants (BFRs):** Combustion of halogenated case material and printed wiring boards at lower temperatures releases toxic emissions including dioxins which can lead to severe hormonal disorders.

(v) **Cadmium**: Cadmium components may have serious impacts on the kidneys. Cadmium is adsorbed through respiration but is also taken up with food. Due to the long half-life in the body, cadmium can easily be accumulated in amounts that cause symptoms of poisoning. Cadmium shows a danger of cumulative effects in the environment due to its acute and chronic toxicity. Acute exposure to cadmium fumes causes flu-like symptoms of weakness, fever, headache, chills, sweating and muscular pain. The primary health risks of long-term exposure are lung cancer and kidney damage. Cadmium also is believed to cause pulmonary emphysema and bone disease (osteomalacia and osteoporosis).

(vi) **Chlorofluorocarbons (CFCs)**: When released into the atmosphere, they accumulate in the stratosphere and have a deleterious effect on the ozone layer. This results in increased incidence of skin cancer in humans and in genetic damage in many organisms.

(vii) **Chromium**: Chromium (VI) is easily absorbed in the human body and can produce various toxic effects within cells. Most Cr (VI) compounds are irritating to eyes, skin and mucous membranes. Chronic exposure to Cr (VI) compounds can cause permanent eye injury, unless properly treated. It may also cause DNA damage.

(viii) **Dioxins:** Dioxins are known to be highly toxic to animals and humans because they bio-accumulate in the body and can lead to malformations of the foetus, decreased reproduction and growth rates and cause impairment of the immune system.

(ix) **Lead**: Short term exposure to high levels of lead can cause vomiting, diarrhea, convulsions, coma or even death. Other symptoms are appetite loss, abdominal pain, constipation, fatigue, sleeplessness, irritability and headache. Continued excessive exposure, as in an industrial setting, can affect the kidneys. It can damage nervous connections and cause blood and brain disorders.

(x) **Mercury:** It is a toxic heavy metal that bio-accumulates causing brain and liver damage, if ingested or inhaled.

(xi) **Polychlorinated biphenyls (PCBs)**: PCBs are known to cause cancer in animals. PCBs are also known to cause a number of serious non-cancer health effects in animals, including effects on the immune system, reproductive system, nervous system, endocrine system and other health effects. PCBs are persistent contaminants in the environment. Due to the high lipid solubility and slow metabolism rate of these chemicals, PCBs accumulate in the fat-rich tissues of almost all organisms (bio-accumulation).

(xii) **Polyvinyl chloride (PVC):** PVC is hazardous because it contains up to 56% chlorine which when burned, produces large quantities of hydrogen chloride gas, which when inhaled, leads to respiratory problems.

(xiii) **Selenium:** Exposure to high concentrations of selenium compounds cause selenosis. The major signs of selenosis are hair loss, nail brittleness, and neurological abnormalities (such as numbness and other odd sensations in the extremities).

Recycling and recovery of e-waste (Metal extraction):

Among the various components found in e-waste, metals significantly contribute to the economy. Hence, efforts are focused on extracting the metals during recycling operation.

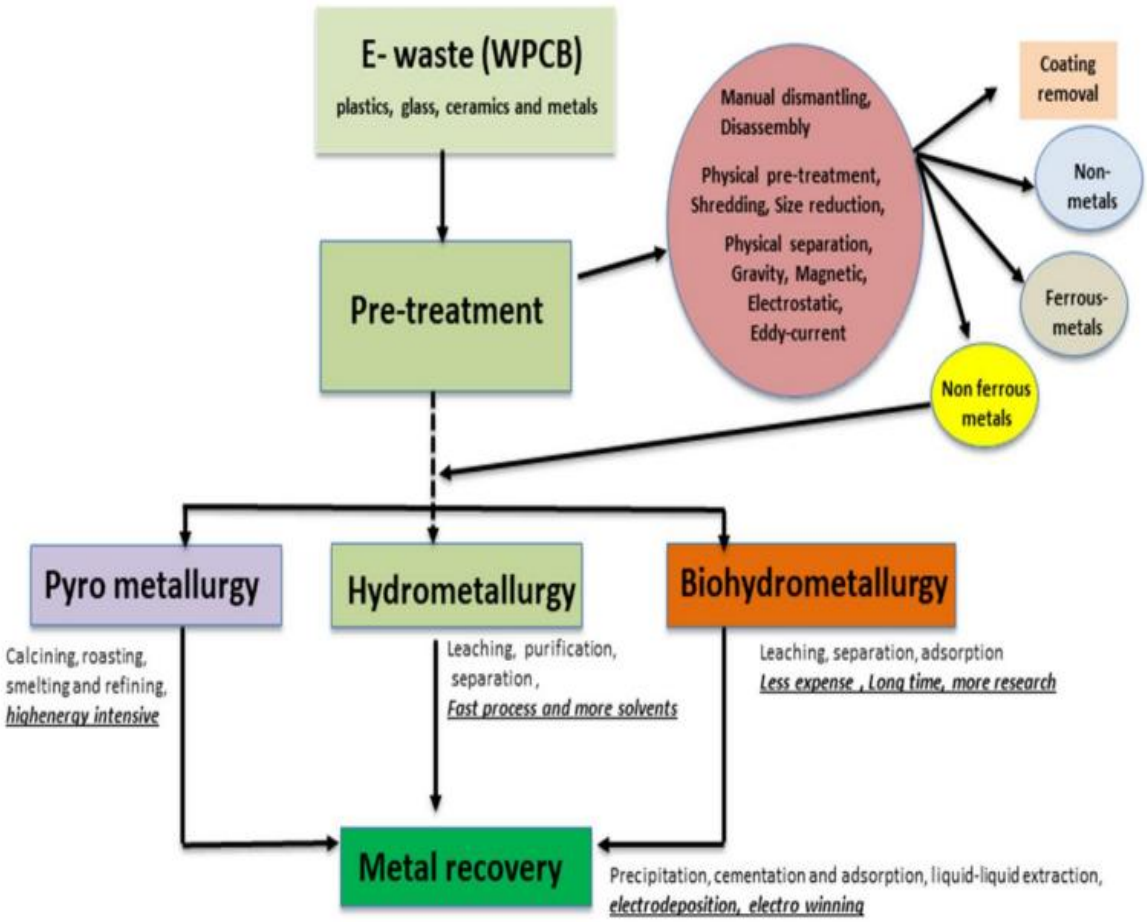

Fig. 2 Overview of stages involved during metal recovery from e-waste

STEP-1: Pre-treatment of e-waste:

The recycling of e-waste is initiated with physical or mechanical pre-process.

1) The first step involves physical removal of toxic materials and unwanted components. Manual dismantling and separation of components such as printed circuit boards, monitors, batteries etc into various fractions like metals, ceramics, plastics, wood and paper using hammer, screwdrivers and conveyer beds for disassembling.
2) In the second step, materials are mechanically shredded through crushers or grinders to collect fragments of metal bearing components. Metals such as Al and Cu can be separated from non-metallic components from eddy current method.
3) Next the waste is passed through electrical separators to separate metallic and non-metallic components.
4) Magnetic separator is used to separate ferrous metals.
5) Gravity separation is used to separate Al metal.
6) Finally, various chemical treatments, such as pyrometallurgical and hydrometallurgical treatments, are employed to recover the precious metals.

STEP-2:

A) Pyrometallurgy

Pyrometallurgical approach of recycling involves thermal treatment of e-wastes to bring about physical and chemical transformations so as to enable recovery of valuable metals. The treatment commonly involves smelting in furnaces at high temperatures, incineration, combustion, and pyrolysis.

(i) Combustion: Combustion and incineration are thermal processes which are adapted to reduce the e-waste volume (up to 90%). Combustion involves burning of e-wastes at high temperature (> 800°C) in presence of oxygen. However, no pollution controlling measures are practiced during open burning, allowing pollutants to escape into the environment. Hence, this method is considered as an environment-invasive process. Although it is a low-cost process suitable for bulk treatment, the formation of toxic gases (dioxins, furans, etc.) during e-waste burning poses severe environmental and health risks.

(ii) Incineration: It is a controlled combustion of e-waste with suitable emission units. The incinerator has two connected furnaces. In the first furnace, e-waste is burnt at temperatures above 800°C. The gaseous products of the first incinerator are further oxidized above 1100°C in the second furnace. Since heavy metals are not typically vaporized at incineration temperatures, these get collected at the bottom of the furnace. Hydrometallurgical processes are used for further recovery of metals such as copper, iron and gold. The emitted flue gases are further treated to remove toxins such as dioxins and acid gases, before releasing it to the atmosphere.

Incineration method involves high capital costs with a risk of air pollution, if the emitted gases are not properly treated.

(iii) Smelting: It is possible to achieve pure Cu recovery (from printed circuit boards) through smelting which involves high temperature (~ 1100 °C) treatment. The processed scrap after

pre-treatment stage contains mainly Fe, Al, Cu, Pb, Sn, Sb, Zn and precious metals. This mixture is fed into copper smelters. During smelting, Pb, Sn, Sb, and other precious metals are collected with the copper parent phase.
It is then casted into anode slab and refined by electrometallurgy. Here, anode is dissolved and pure copper (99.9%) is deposited at cathode, leaving behind the slurry at the anode. The slurry, also known as cathodic slime, is rich in precious metals such as Pt, Sb, etc. They are recovered later by hydrometallurgy.
Copper smelting is ideal for recovery of non-ferrous metal fractions from e-waste. However, the process is highly energy-intensive, requiring high temperatures (300–1100 °C), and can also generate toxic gases involving corrosive acids like H_2SO_4, HCl, and HNO_3.

(iv) Pyrolysis: Pyrolysis is a thermal decomposition process carried out in the absence of oxygen, typically at 300 - 800°C. Unlike combustion or incineration, pyrolysis does not burn the material; instead, it breaks it down into gases (e.g. methane, hydrogen), liquid (pyrolysis oil which can be used as fuel) and solid residue. The solid residue or char is rich in metals which can be further treated by suitable hydrometallurgical processes to extract precious metals such as copper, gold, silver, aluminium and palladium. Printed circuit boards can be treated in a pyrolysis reactor to extract copper and gold.
Pyrolysis method demands expensive set-up. Yet is considered as a cleaner, safer and more efficient thermal process.

B) Hydrometallurgy
Hydrometallurgical process is chemical-based method to extract valuable metals from e-waste. It involves leaching of metals from pre-treated solid e-waste in to liquid solution using acids or other chemical agents. It may also contribute as a secondary process for extracting metals from pyrometallurgical solid residue. The metal aqueous solution requires further separation and purification to obtain isolated metals.
The common leaching agents are

(i) Acid Leaching*:* Nitric acid, sulphuric acid and hydrochloric acid are used to leach the target metals such as copper, zinc and gold. Organic acids such as ascorbic acids, citric acids and acetic acid are used to leach metals, such as lithium, from spent batteries and mobile devices.
Demerit: Use of strong acid generates a corrosive media.

(ii) Cyanide leaching*:* Precious metals such as gold and silver are recovered using cyanide leaching. Here, the metals are treated with a dilute solution of cyanide (KCN/NaCN) to dissolve the metal as metal-cyanide complex.
Demerit: The process produces toxic cyanide-containing wastewater which requires strict disposal measures.

(iii) Non-cyanide leaching: Due to the toxic nature of cyanide, non-cyanide leaching agents such as thiourea, thiosulphate and halides are being preferred as a non-toxic alternative for hydrometallurgical processes.

- **Thiourea leaching***:* Thiourea (NH_2-CS-NH_2) is used as a complexing agent that forms cationic soluble complex with target metal, typically gold. Thiourea is not stable and decomposes easily in alkaline solutions and hence the reaction is carried in acidic conditions in presence of oxidizing agents such as hydrogen peroxide (H_2O_2).
  Demerit: Gold-thiourea complexes are less stable than gold-cyanide complexes making the metal recovery less efficient.
- **Thiosulphate leaching:** A solution containing thiosulphate ($Na_2S_2O_3$ or $(NH_4)_2S_2O_3$), copper ions and ammonia is used as a leaching solution to dissolve the metals in to soluble thiosulphate complexes. Copper ions act as catalyst while ammonia helps in stabilizing the metal-thiosulphate complexes.
  Demerit: Higher concentration of thiosulphate is required for efficient leaching. Recovery of metal from thiosulphate solution is more energy-consuming than from cyanide solution.
- **Halide leaching:** This type of leaching utilizes halide ions (chloride, bromide and iodide), in presence of oxidizing agents such as Cl_2 gas or H_2O_2, to convert metals into soluble halide complexes.
  Demerit: Chlorine, bromine and iodine are corrosive chemicals and require treatment before disposal.

STEP 3: Metal Recovery

In this process, metal is recovered from leached metal solution. A variety of methods such as solvent extraction, ion exchange, adsorption and electrodeposition are followed.

(i) Solvent extraction*:* Solvent extraction involves selectively transferring metal ions from one liquid phase (mostly aqueous) to another phase (organic). Initially, leachate is treated with organic solvent containing metal-specific extractant. For example, amides or amines are used for extraction of V, Ir, Rh and W; Methyl iso butyl ketone is used to extract Au; Diamine is used to extract Pt and Pd. The metal ions get transferred to the organic phase. The organic phase is now separated and further treated with stripping solution (acid or alkali) to transfer the metal ions back to the aqueous phase. The metal-free organic solvent is re-used while the metal-rich aqueous phase is separated and further processed to recover the pure metal.

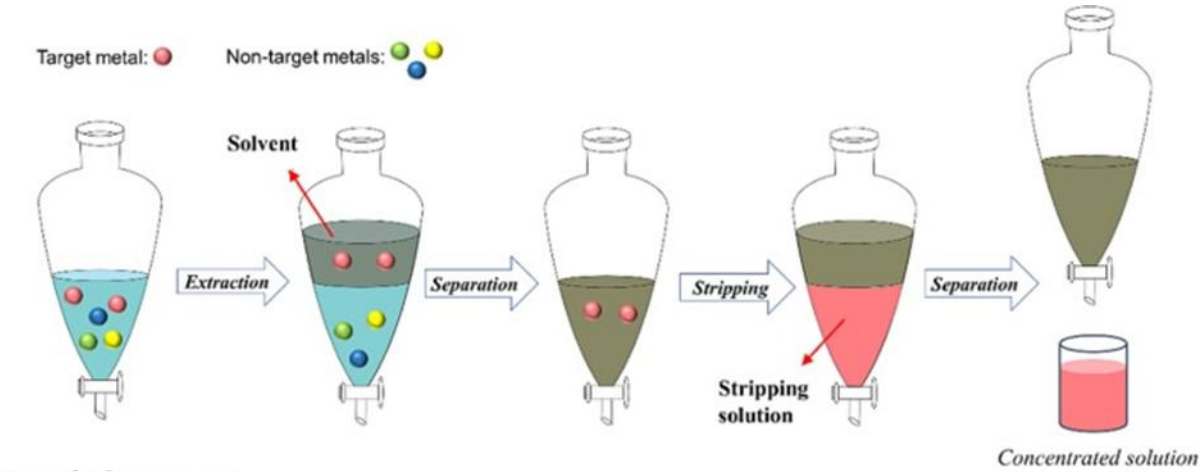

Fig. 3 Metal recovery from solvent extraction process

(ii) Ion exchange: Ion exchange process can be used to recover metals from the leach solution by selectively binding the metal ions to a solid resin (polymer beads). Post-leaching, the metal ion containing leach solution is passed through a column filled with ion exchange resin. The functional groups of the resin selectively bind metal ions by exchanging the metal ions with the resin cations. Thus, unwanted impurities are washed out while the resin retains the metal ions. The bound metal ions can be stripped off by treating the resin with eluant (solution that has a higher affinity for metal ions), causing the metal ions to be released from the resin. The eluant is collected and further treated to recover the pure metal. In the process, the resin is regenerated for further use. E.g. Anion exchange resins are used to extract gold from $[AuCl_4]^-$ leachate while cation exchange resins are used to recover Cu, Ni and Co.

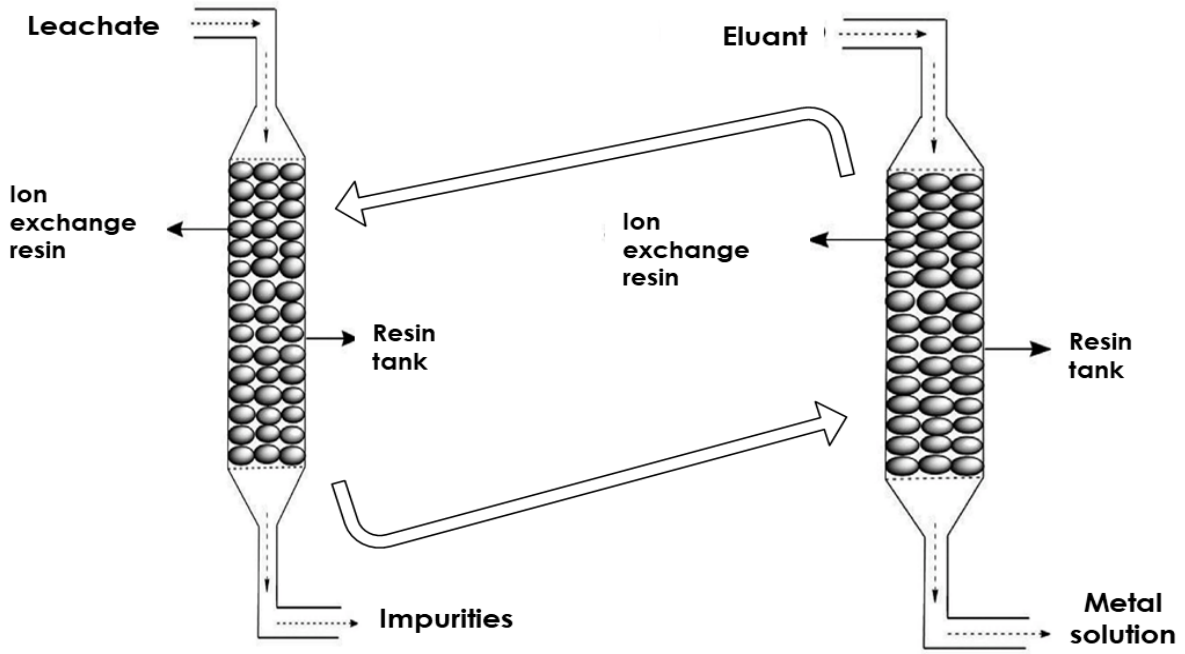

Fig. 4 Metal extraction through ion exchange process

(iii) Adsorption*:* Metal recovery through adsorption is a surface-bound process where the metal ions from a solution adhere to the surface of a solid adsorbent. The adsorption may take place through physical process (weak electrostatic forces) or chemical process (chemical bonding). The adsorbed metal ions can be effectively separated through desorption process, using a specific solvent or by varying the pH, thereby allowing the adsorbent to be reused.

(iv) Electrodeposition: Electrodeposition is an electrochemical process used to recover metals from solutions by depositing them on to a cathode. The leachate or the recovered concentrated solution, which is rich in metal ions, is taken as the electrolyte. Pure metal same as metal to be extracted is taken as cathode and inert metal as anode. On applying current, the metal ions are reduced and pure metal gets electrodeposited at cathode. Metals such as gold, lead, tin, copper can be recovered through this method. Metals with purity as high as 99% can be obtained. It is also a highly efficient and environmentally safer method.

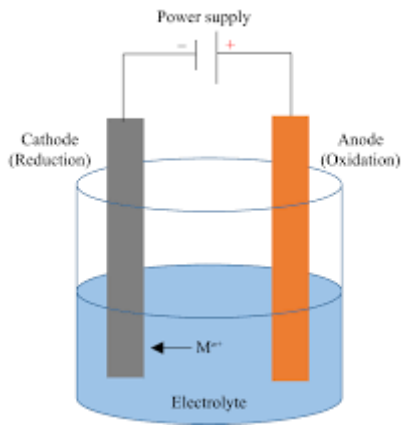

Fig. 5 Electrodeposition of pure metal on cathode using metal-rich electrolyte

Extraction of gold from e-waste:

Gold has good electrical conductivity and chemical stability and hence it is used for making integrated circuits of electronic devices, coating for contacts and connectors. E-waste contains a significant amount of gold, often higher than those found in natural ore deposits. Also, the process of extracting gold from e-waste is more cost-efficient than traditional mining. On an average, 1.5 kg of gold can be extracted from one ton of e-waste.

Gold Recovery (Hydrometallurgical method)

There are three stages in recovery of gold by hydrometallurgical method

(i) Pretreatment stage

(ii) Chemical treatment stage

(iii) Metal recovery stage

(i) Pretreatment stage

In the pretreatment stage, e-waste is manually dismantled to separate various fractions like metals, ceramics, plastics, wood and paper. Techniques such as gravity separation electrostatic separation, magnetic separation and eddy current separation are used to separate metals from fractions.

(ii) Chemical treatment stage

Two different leaching agents can be used to effectively leach gold. They are aqua regia leaching and cyanide leaching.

a) Aqua regia leaching: Aqua regia is mixture of nitric acid and hydrochloric acid in 3:1 ratio. It dissolves gold into chloroauric acid.

$$Au + HNO_3 + 4HCl \rightarrow HAuCl_4 + NO + 2H_2O$$

b) Cyanide leaching: Sodium salt of 3-nitrobenzene sulfonic acid with potassium cyanide in the presence of oxygen is used as leaching agent. A water soluble dicyanoaurate gold complex is obtained in the process.

$$4Au + 8CN^- + O_2 + 2H_2O \rightarrow 4Au(CN)_2^- + 4OH^-$$

(iii) Metal recovery stage

In the last step, metal is recovered from the leach solutions. Various methods such as electrodeposition, solvent extraction, ion exchange, adsorption and chemical precipitation are used to recover gold. In a typical electrodeposition method, pure gold metal (cathode) and inert anode are dipped in the leach solution. When current is applied, pure gold is selectively electrodeposited on the cathode, which is later recovered through melting.

Extraction of copper from e-waste:

Copper is one of the most important metals in e-waste. Copper is a non-ferrous metal with unique properties such as good ductility, high thermal and electrical conductivity. It is widely used in various industries and is a significant part of printed circuit boards. A typical board can contain around 5 - 35% Cu by weight. Due to lack of copper resources, it becomes essential to extract copper from these wastes.

Copper Recovery (Hydrometallurgical method)

There are four stages in recovery of copper by hydrometallurgical method

(i) Pretreatment stage

(ii) Chemical treatment stage

(iii) Purification stage

(iv) Metal recovery stage

(i) Pretreatment stage:

In the pretreatment stage, e-waste is manually dismantled to separate various fractions like metals, ceramics, plastics, wood and paper. Techniques such as gravity separation electrostatic separation, magnetic separation and eddy current separation are used to separate metals from fractions.

(ii) Chemical treatment stage

The leaching agents to dissolve copper can be acidic or alkaline. In acidic conditions, copper can be efficiently dissolved in to its corresponding salt in the presence of oxidizing agent.

$$Cu + H_2SO_4 + H_2O_2 \rightarrow CuSO_4 + 2H_2O$$

(iii) Purification stage (Solvent extraction)

The strong acidic conditions during leaching results in the dissolution of other undesired metals such Fe, Zn or Pb along with Cu. In order to purify the leachate and increase the concentration of copper ions, the leachate is subjected to solvent extraction.

The aqueous phase (containing copper ions) is mixed with an organic solvent containing an extractant (oxime derivatives). At a pH of 2, the extractant binds to Cu^{2+} ions, thus transferring them in to the organic phase. The organic phase is then collected and treated with strong acidic solution. Copper ions get released from the organic phase back into the new aqueous phase. The impurity-free and copper-rich aqueous solution is used as an electrolyte for selective recovery of copper through electrodeposition.

(iv) Metal recovery stage

During electrodeposition, stainless steel or pure copper sheet is taken as cathode and titanium or lead alloy rod is taken as the inert anode. The two electrodes are dipped in the acidic copper-rich electrolyte. When current is applied, copper ions are selectively reduced and get deposited on cathode. Pure copper can be obtained by peeling off or melting the copper layer deposited on the cathode substrate.

Role of stakeholders in environmental management of e-waste

The following stakeholders play a crucial role in environmental management of e-waste. They are as follows.

1. Statutory Government Regulatory bodies
2. Producers (Manufacturing units)
3. Recyclers (Recycling units and collection units)
4. Consumers

All stakeholders must effectively work in tandem form for effective e-waste management.

1. Role of Statutory Government Regulatory bodies

The e-waste management program is designed by statutory government regulatory bodies. The members of the body, frame the policies and execute it for protection of the environment. In India, the key statutory bodies include the Ministry of Environment, Forest and Climate Change (MoEFCC), Central Pollution Control Board (CPCB), and State Pollution Control Boards (SPCBs).

(i) Formulation of Policies and Regulations: Statutory bodies create legal frameworks to manage e-waste by setting standards for collection, recycling and disposal of electronic goods. The CPCB and SPCB play crucial role in implementing these frameworks at the central and local levels. The E-waste Management Rules established in 2011, was amended in 2018, to promote sustainable e-waste management practices throughout the nation.

(ii) Authorization and Licensing: SPCBs authorize e-waste recyclers and dismantlers, ensuring they have the necessary infrastructure and follow environmental safety standards. Only certified and compliant entities are authorized to handle e-waste.

(iii) Enforcement and Monitoring: Statutory bodies monitor the entire e-waste management system, including collection, transportation, and disposal. They investigate complaints, conduct inspections, and take necessary action against those who violate e-waste management rules.

(iv) Promoting Extended Producer Responsibility (EPR): Statutory bodies implement EPR schemes requiring producers to take responsibility for collection and recycling of their products. They also encourage manufacturers to design eco-friendly and recyclable electronics.

(v) Public awareness and Education: Statutory bodies are responsible for raising public awareness about e-waste management and promoting responsible disposal practices among consumers and producers.

(vi) Facilitating Research and Development: Statutory bodies play a crucial role in encouraging and promoting R&D in safer recycling technologies and environment friendly disposal methods.

(vii) Coordination with other agencies: Statutory bodies cooperate with local bodies and NGOs to tackle the e-waste management issues at the local level. At the same time, it also works with international bodies to manage transboundary movement of hazardous wastes.

2. Producers (Manufacturing Units)

Producers, including electronics manufacturers and brand owners, play a crucial role in minimizing the environmental impact of e-waste by fulfilling the following roles:

(i) Extended Producer Responsibility (EPR): Every manufacturer has a legal obligation under the E-waste Management Rules to take responsibility for the end-of-life management of their products, including collection, recycling and proper disposal. The producers can do so by establishing collection centres, take-back schemes or even providing reverse logistics to retrieve old products from consumers. They can also take the help of Producer Responsibility Organizations (PROs) which act as intermediates, helping producers meet their EPR obligations efficiently.

(ii) Sustainable Product Design: Manufacturers should design products considering the longevity and recyclability of the device. Minimization or elimination of hazardous and toxic materials in their products is another important aspect of sustainable design.

(iii) Raising Consumer Awareness: Manufacturers must inform the consumers about the impact of the hazardous materials through proper labels. The environmental effects of unsafe e-waste handling must be conveyed to the consumers through campaigns or official websites.

(iv) Reporting and Compliance: Producers must maintain and submit records of e-waste collected and processed. They must comply with all relevant e-waste management regulations.

3. Recyclers

Recyclers play a crucial role in the responsible management of e-waste by dismantling, recycling, and safely disposing electronic components. They ensure that hazardous materials are managed properly, preventing contamination of soil and water, and also recover valuable materials like gold, silver, and copper for reuse.

(i) Collection, Dismantling and Storage: The recyclers play a role in collecting the e-wastes from collection units, dealers or also directly approach consumers for door-to-door collection. The recyclers must use approved technologies and methods to treat and recycle e-waste without releasing toxic gases to the environment. At the same time, they must ensure personal safety during manual dismantling and segregation of metals, glass and other hazardous materials. They are also responsible for proper storage of hazardous components to prevent leaks and fires.

(ii) Waste Disposal: Recyclers must manage and dispose off any residue or waste generated during the recycling process in authorized Treatment, Storage and Disposal Facilities (TSDFs). They should have adequate facilities for managing spills and leaks of hazardous substances, such as compressor oils and refrigerants.

(iii) Capacity Building: Recyclers must train workers on safe handling practices and environmental safety. They need to collaborate with producers under EPR schemes and NGOs for safe collection and processing of e-waste.

(iv) Registration: The recyclers must have obtained a valid consent to operate by registering themselves with the SPCB and accordingly adhere to the guidelines and standard set by the

CPCB. Recyclers have to cooperate with the government bodies during the periodic audits and inspection by providing details about their involvement in EPR activity, amount of e-waste processed and valuable materials recovered.

4. Consumers

Consumers play a vital role in the proper environmental management of e-waste. They are responsible for segregating e-waste, utilizing authorized collection centers, avoiding unauthorized disposal and considering reuse and donation options before disposal.

(i) Segregation: Consumers should separate electronic waste from regular household trash to ensure it is handled correctly. In case the electronic device contains any private or important data, it is the responsibility of the consumer to delete all personal data file to ensure protection of privacy and security.

(ii) Authorized collection: E-waste should be dropped off at designated collection points or handed back through manufacturer take-back programs, rather than being discarded with regular waste. Consumers must avoid selling e-waste to informal or unregulated handlers who use unsafe methods for e-waste management.

(iii) Reduce, Reuse and Recycle: Consumers should limit electronic purchases by avoiding purchase of unnecessary and excessive electronic items. This is the major way to reduce e-waste generation. Consumers should consider the lifespan and recyclability of electronics when making purchasing decisions. They must use electronics responsibly to extend their lifespan. If possible, the devices must be repaired or donated instead of discarding them early. They must see to it that the e-waste are recycled scientifically.

(iv) Awareness: Consumers must be aware of the different laws and regulations on e-waste disposal. They must also have a knowledge of the environmental hazards due to improper e-waste disposal. They should encourage others to practice responsible e-waste disposal and identify the recycling options in their locality.

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